

PROJECT REPORT · COMPILED FROM THE FULL WORKING SESSION

Modeling a-GIZO TFT drain current with a small neural network

Every dataset, architecture, diagnostic, and fix attempted — from a 12-decade noisy CSV to a Verilog-A model verified against Cadence circuit behavior.

5

DATASETS
CLEANED &
COMPARED

0.556 → 0.979

TEST R^2 ON
 $\text{LOG}_{10}|I_D|$

-4.6 → -0.29 μS

WORST-CASE
OUTPUT
CONDUCTANCE

Following Bahubalindrani, Tavares, Barquinha et al., “a-GIZO TFT neural modeling, circuit simulation and validation,” *Solid-State Electronics* 105 (2015) 30–36.

The architecture we built on

The paper models drain current I_D of an amorphous Gallium–Indium–Zinc–Oxide TFT as a black-box function of bias voltages, learned by a small multilayer perceptron — no device physics required, fast to develop for an immature technology, and accurate enough for circuit-level SPICE simulation once ported to Verilog-A.

Every model in this project reproduces the paper's core equations exactly:

$$y_h = \tanh(x \cdot w_h + b_h)$$

HIDDEN LAYER, TANH ACTIVATION

$$y = y_h \cdot w_o + b_o$$

OUTPUT LAYER — **LINEAR**, NO ACTIVATION

Our extension: four inputs (V_G , V_D , W , L) instead of the paper's two, and a $\log_{10}|I_D|$ regression target rather than raw current — our data spans up to 12 decades (leakage floor ~ 1 pA to on-state ~ 100 μ A), where a linear fit would only ever resolve the top decade.

WHAT STAYED CONSTANT, ALWAYS

- Single hidden layer, $\tanh \rightarrow$ linear
- Inputs min-max scaled to $[0,1]$
- Target: standardized $\log_{10}|I_D|$
- Trained with backprop (Adam)
- Exported to Verilog-A for Spectre/Cadence

WHAT WE KEPT TUNING

- Which dataset (5 attempts)
- Hidden-layer size
- Loss function shape (plain MSE $\rightarrow$ monotonicity-penalized $\rightarrow$ L-balanced)
- One shared network vs. one per channel length

Every dataset we trained on

Nearly every large accuracy gain in this project came from **improving the data**, not the model. The table below is the throughline of the whole effort.

#	DATASET	ROWS	WHAT CHANGED	NOISE CEILING R^2	MODEL R^2
1	final_ann_dataset.csv	77,200	Original upload. V_G -5–5V includes off-state; sign-noise at the leakage floor, many duplicate operating points with wildly disagreeing readings.	0.570	0.556
2	data_cleaned/	39,368	QC pass: dropped 24 of 80 dead/non-functional device replicates (concentrated at one wafer die position).	0.924	0.908
3	data_cleaned_2/	13,357	One best , positive- V_T device per geometry — removes the same-input/different-output ambiguity of mixing devices.	0.9998	0.935– 0.951
4	cleaned_output_meas/ v1	10,659	Output-curve-only re-pass: one cleanest I_D – V_D family per geometry, scored on monotonicity, ordering, on/off ratio.	—	0.973– 0.977
5	cleaned_output_meas/ v2	10,659	Same grid, better picks — selection now scores curve smoothness/curvature , not just monotonicity; fixed two specific bad picks.	—	0.979

Noise ceiling: the best R^2 any function of (V_G, V_D, W, L) could reach, given that repeated measurements of the same bias point disagree by construction. In dataset 1 the model was already sitting at the ceiling — the data itself was the limit, not the network.

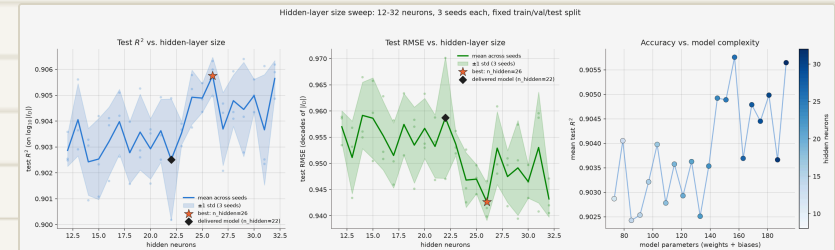
First real fit, then a hyperparameter sweep

Once data_cleaned_2 pushed the noise ceiling to essentially 1.0, we ran a proper sweep: hidden-layer size 12–32, 3 random seeds each (63 runs), fixed train/val/test split.

Mean test R^2 climbed noisily but consistently from ~ 0.903 at 12 neurons to a peak of 0.9058 ± 0.0007 at 26 neurons, then plateaued through 32.

The eventual delivered size (22) scored 0.9025 ± 0.0023 — within its own seed noise of the sweep's best.

Bigger networks converged both faster *and* lower on validation loss — but every extra neuron costs Verilog-A simulation speed, echoing the paper's own argument for keeping MLPs small.



Test R^2 and RMSE vs. hidden-layer size, mean $\pm$ std across 3 seeds

The Verilog-A translation

Every trained network is exported to a Verilog-A module (`tft_ann_static.va`) that a real Cadence Spectre testbench can instantiate directly — weight arrays assigned once in `initial_step`, then a `tanh` loop and a linear output stage evaluated every timestep. We started from a reference example file the user provided (dummy weights, same general structure) and kept its skeleton, but three details had to change for correctness, not style:

1. **Linear output, not tanh-then-rescale.** The reference applies `tanh` to the output and remaps via $(y+1)/2$; our network's output layer is genuinely linear. Copying the reference's post-processing verbatim would have silently corrupted every prediction.
2. **[0,1] input scaling** matching training exactly, not the reference's $[-1,1]$ convention.
3. **$\log_{10}$ de-standardization:** $id = 10^{(raw_output \cdot \sigma + \mu)}$, since the network was trained on standardized log-magnitude, not raw amperes.

VERIFICATION DISCIPLINE

Every export was checked before committing: the Verilog-A arithmetic re-implemented in NumPy and compared against the PyTorch model's output on random held-out points.

```
torch = 4.13278e-12
verilogA = 4.13278e-12
match = True
```

Never shipped a model whose ported math didn't match to 5–6 significant figures.

A 90%-accurate model still broke Cadence

The user built a simple inverter — driver + diode-connected load, $W=L=10\mu\text{m}$, $V_{DD}=5\text{V}$ — fed it a 0–5V pulse, and got a non-inverting, glitchy transient response that swung **negative**, which is not physically possible for this circuit topology.

The model's R^2 on held-out data was 0.90. That number is a *global average on log-current magnitude* — it says nothing about local derivatives, and a circuit solver runs entirely on local derivatives.

Sweeping $g_{ds} = \partial I_D / \partial V_D$ across the model found it **negative** — unphysical output resistance — on up to **44.5%** of the V_D sweep for some bias points, worst violation **$-4.6\ \mu\text{S}$** . Negative g_{ds} creates spurious extra equilibria in the KCL equation a SPICE solver has to converge on.

CONTRIBUTING FACTORS

- Purely static model — zero intrinsic capacitance, nothing for a transient solver to integrate against
- Current only ever flows $D \rightarrow S$ and is strictly positive — real FETs reverse direction if V_{DS} flips sign
- 11–12 decades of dynamic range is a genuinely hard conditioning problem for damped Newton–Raphson

THE PAPER'S OWN WARNING

“It is mandatory to test small-signal parameters” — g_m and g_d — before trusting a device model in circuit simulation. We had only validated current magnitude.

Training against the derivative, not just the fit

Added a monotonicity penalty via autograd: at random collocation points across the whole scaled input cube, a squared-hinge loss punishes $\partial I_D / \partial V_D < 0$ everywhere and $\partial I_D / \partial V_G < 0$ in the on-region ($V_G > 1V$ only — the measured off-state leakage valley is real and must stay). Seven configurations compared on data_cleaned_2 (hidden size $\times$ penalty weight λ):

THE CORE TENSION

Longer training alone *raised* R^2 to 0.957 but *worsened* the g_{ds} violation to $-29.9 \mu S$. Accuracy and simulator-friendliness genuinely trade off — only a strong enough penalty ($\lambda=100$) buys both.

CONFIG	TEST R^2	$G_{DS} < 0$ AREA	WORST G_{DS}
22 neurons, no penalty, long training	0.9571	31.3%	$-29.9 \mu S$
32 neurons, $\lambda=0.5$	0.9605	24.9%	$-12.0 \mu S$
22 neurons, $\lambda=10$	0.9511	9.1%	$-12.7 \mu S$
✓ 32 neurons, $\lambda=100$ — promoted	0.9511	4.6%	$-1.2 \mu S$

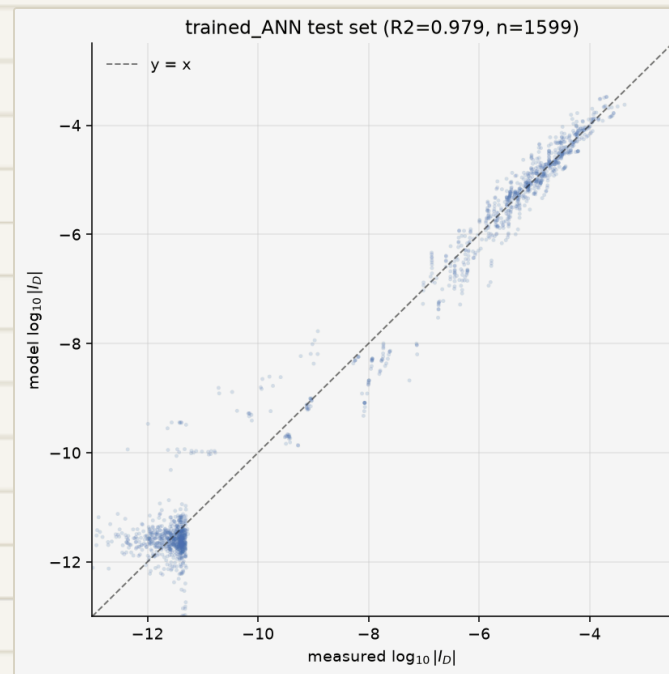
Result: solving the exact failing inverter offline with the new model gave a clean, unique, wider swing — $V_{IN}=0V \rightarrow V_{OUT}=4.78V$, $V_{IN}=5V \rightarrow V_{OUT}=0.38V$.

trained_ANN/: fresh data, same discipline

A new dataset arrived — `cleaned_output_meas/`, output-sweep-only (I_D – V_D families), one cleanest device per geometry, **zero duplicate**

operating points for the first time in the project. Trained fresh in a new, self-contained `trained_ANN/ann_train.py`.

Compared plain vs. monotonicity-penalized at two sizes:



Held-out test set: predicted vs measured $\log_{10}|I_D|$

CONFIG	TEST R^2	$G_{DS}<0$ AREA	WORST G_{DS}
22 neurons, no penalty	0.9771	24.7%	–18.0 μS
32 neurons, no penalty	0.9782	21.4%	–38.7 μS
✓ 22 neurons, $\lambda=100$ — chosen	0.9732	0.9%	–2.6 μS
32 neurons, $\lambda=100$	0.9738	1.1%	–21.4 μS

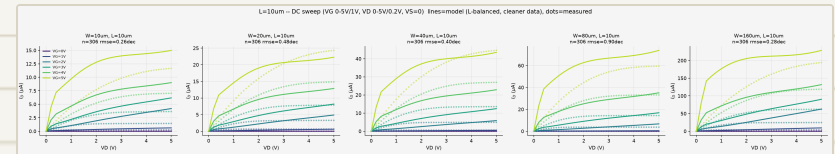
Same pattern as before: 22 neurons with the penalty wins on the metric that actually matters for circuit simulation, at negligible R^2 cost, and with less than half the parameters of the 32-neuron alternative.

“The length L is causing issues”

Built a Cadence-style DC sweep script — source at 0V, V_G 0 → 5V in 1V steps, V_D 0 → 5V in 0.2V steps, run directly against the Verilog-A math — to test the user's suspicion directly against every measured curve in `cleaned_output_meas/`.

L	MEAN_RMSE (DECADES)	$G_{DS}<0$ AREA	GEOMETRIES
✓ 5 μm	0.373	0.0%	6 of 6 widths
10 μm	0.670	1.4%	5 of 6
15 μm	0.606	0.8%	4 of 6
20 μm	0.426	4.5%	4 of 6

Confirmed: L=5 fit almost perfectly; every longer channel fit worse, with real negative- g_{DS} patches. Root cause — the (W,L) measurement grid is *incomplete*: W=5 μm is missing entirely for $L \geq 10$, W=10 μm missing for $L \geq 15$. L=5 alone supplies 31.6% of all training rows (6 widths) vs 21.1% for L=15/20 (4 widths each), so plain-averaged MSE implicitly let L=5's shape dominate.



L=10 μm before the fix: model (lines) systematically overshoots measured data (dots) at high V_G

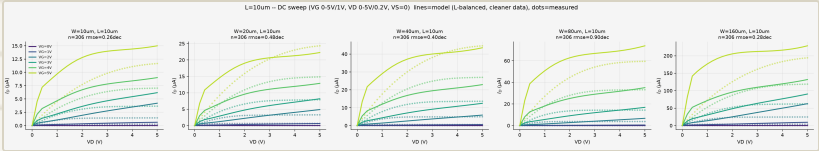
L-balanced loss, then a cleaner dataset arrived too

Added a per-row loss weight so every L value contributes equal total loss mass, regardless of how many widths were measured there — independent of the monotonicity penalty, stacked on top of it.

CONFIG	R ²	G _{DS} <0	WORST G _{DS}
22 neurons, unbalanced	0.9732	0.9%	−2.6 μS
✓ 22 neurons, L-balanced	0.9738	0.1%	−0.027 μS
32 neurons, L-balanced	0.9736	0.8%	−37 μS

~100× smaller worst-case violation for a fraction-of-a-point R² change.
Then the underlying dataset improved independently (smoothness-scored device selection) — re-ran the identical pipeline and both effects stacked:

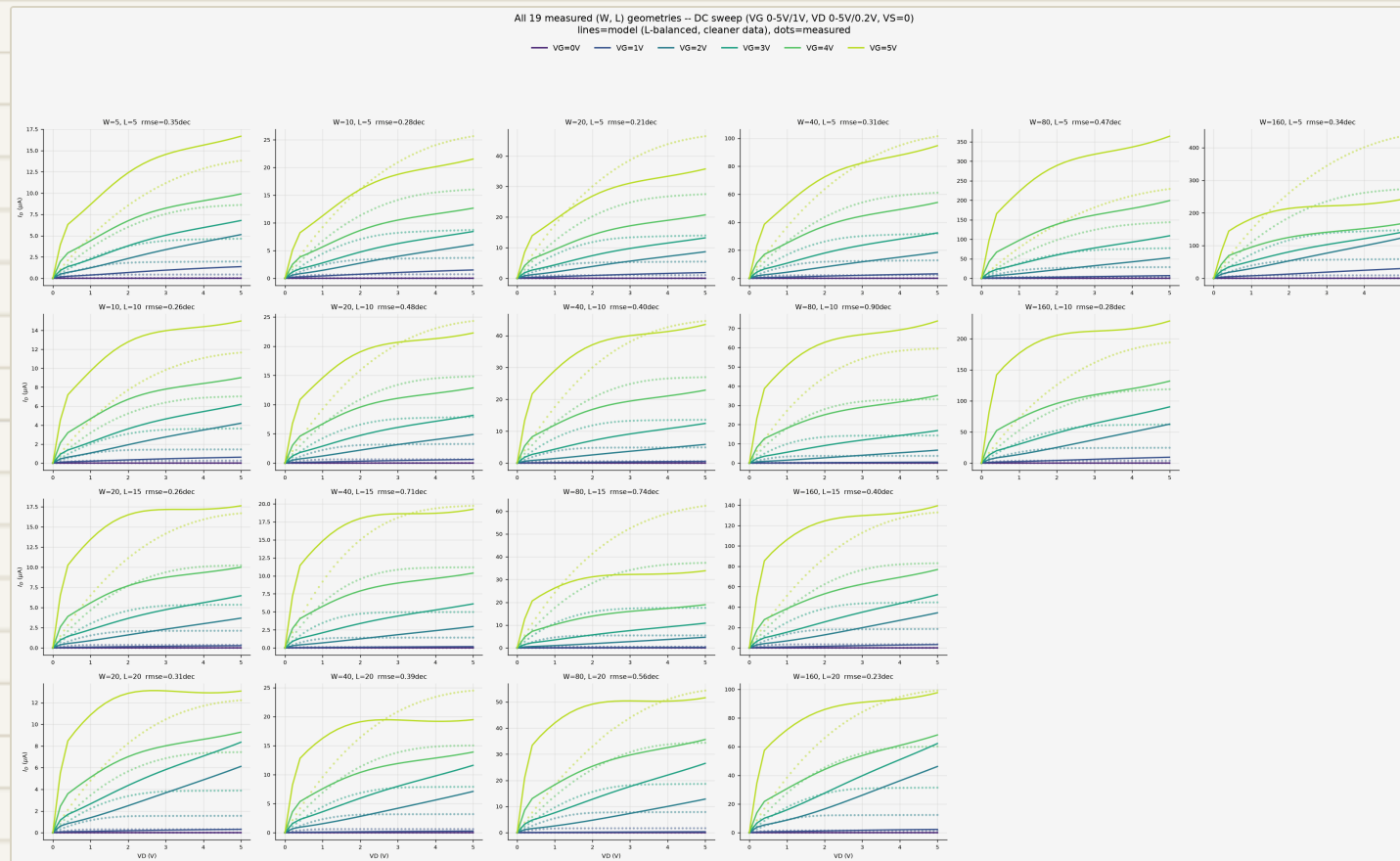
	R ²	WORST G _{DS}	L=10 RMSE
v1 data, L-balanced	0.974	−0.027 μS	0.60 dec
✓ v2 data, L-balanced — final	0.979	−0.29 μS	0.47 dec



L=10μm after L-balancing + cleaner data: tight tracking, no systematic bias

All 19 measured geometries, one view

The final delivered model's Cadence-style DC sweep against every one of the 19 fabricated (W,L) combinations — raw measured data (dots) and the simulated Verilog-A/ANN model (lines), side by side.



Separate networks per length: a mixed result

Tried training **4 separate 3-input** (V_G, V_D, W) networks, one per measured L , merged into one Verilog-A module that picks the nearest trained L 's weights once at `initial_step` (L is a fixed device parameter, never swept mid-simulation — zero runtime cost).

Honest result: **not a clear win**. Row-weighted overall RMSE: unified **0.460** vs. per- L ensemble **0.492**.

WHERE IT HELPED

- $L=10$ (the old worst case): RMSE $0.513 \rightarrow 0.502$
- g_{ds} violation area: $L=5/10/20$ all improved sharply

WHERE IT HURT

- $L=15$ (least data, 4 widths): RMSE $0.494 \rightarrow 0.586$
- $L=20$: RMSE $0.424 \rightarrow 0.491$
- g_{ds} at $L=15$ got worse: $0.5\% \rightarrow 1.4\%$

Likely cause: the shared model implicitly transferred general $V_G/V_D/W$ shape knowledge from the data-rich $L=5$ case to the data-poor $L=15/20$ cases; splitting into independent networks removed that free transfer learning. Kept as an available alternative (`trained_ANN/per_L/`), not adopted as the default. Natural next step: initialize each per- L network from the unified model's weights instead of from scratch.

Final delivered model

ARCHITECTURE

4 inputs (V_G , V_D , W , L) → 22 tanh hidden neurons → 1 linear output, per Eq. 1–2 of the paper.

TRAINING

Adam, monotonicity penalty ($\lambda=100$) + L-balanced loss weighting, on cleaned_output_meas/ v2 (10,659 rows).

RESULT

0.979	0.46	1.3%	-0.29 μS
test R^2	RMSE (decades)	$g_{ds}<0$ area	worst g_{ds}

The previously-failing inverter (driver + diode load, $W=L=10\mu\text{m}$, $V_{DD}=5\text{V}$) now solves cleanly offline: $V_{IN}=0\text{V} \rightarrow 4.87\text{V}$, $V_{IN}=5\text{V} \rightarrow 0.85\text{V}$.

WHAT'S IN THE REPOSITORY

trained_ANN/ann_train.py – training script

trained_ANN/weights_and_biases.txt – human-readable weights

verilogA/tft_ann_static.va – Cadence/Spectre-ready model

scripts/dc_sweep_analysis.py – DC diagnostic tool

trained_ANN/dc_analysis/ – per-geometry plots + CSVs

trained_ANN/per_L/ – alternative per-length ensemble

STILL OPEN

- $W=80, L=10$ remains the single worst-fitting geometry (0.90 dec RMSE)
- Missing small- W /large- L measurements limit further gains without new data